

Membership inference identifies concentrated re-identification risk in synthetic continuous glucose monitoring data

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Abstract

Synthetic continuous glucose monitoring data is increasingly deposited in public repositories to enable sharing, on the premise that a release containing no copied record cannot identify anyone. We tested that premise directly. Using a public multi-study cohort of 506 people, we trained a generator on a background population and 26 further generators differing from it by exactly one patient, and asked whether that patient’s real recordings lay closer to the release that had seen them. Exposure was bounded between a generator that memorises by construction and a permutation floor computed separately for each released dataset, which is not near zero. Of seven generators, the strongest exposed 24 of 26 patients against a floor of seven, and all thirteen matched controls were among them: from glucose alone, exposure is close to universal, and patients with atypical profiles are distinguished by the confidence of their recovery rather than by its occurrence. Two coordinates carry the entire recoverable signal, absolute level and temporal ordering, and removing both returns exposure to the floor. A generator whose latent bottleneck cannot store level per timepoint reaches that floor at the best fidelity we measured, and the same coordinate can be removed at release time by a custodian who cannot retrain, leaving translation-invariant glycaemic metrics exactly unchanged.

Keywords: synthetic data, membership inference, continuous glucose monitoring, privacy, generative models, type 1 diabetes

1 Introduction

Continuous glucose monitoring produces one of the densest records in routine care: a value every five minutes, indefinitely, for millions of people living with diabetes. It is also among the hardest records to share. A trace carries sleep, meals, work patterns and illness alongside glycaemia, and the consent under which it was collected rarely anticipates secondary use. Synthetic data is the most widely proposed way out of that bind — release a generated cohort that reproduces the statistics of the real one and share it without restriction — and the proposal is now moving from argument to obligation, with journals and funders increasingly requiring that a synthetic dataset be deposited publicly, with a persistent identifier and an open licence, as a condition of publication. TODO — citations for the clinical value of CGM, for synthetic data as a sharing route, and for the deposition requirements.

The privacy of such a deposit is usually argued rather than measured. The common arguments are that no real record appears in the release, or that the distance from each synthetic record to its nearest real neighbour exceeds some threshold. Neither is a statement about what an adversary can infer. Membership inference — asking whether a particular person’s data was in the training set — is the standard formulation of that question, and it has been studied extensively for classifiers and for images, but far less for generative models of physiological time series, where the attacker sees only released samples and never the model. Continuous glucose monitoring has attracted particularly little attention, partly because glucose is homeostatically bounded and a trace is widely assumed to be too constrained to identify anyone. TODO — citations for membership inference, for its application to generative models, and for existing privacy evaluations of synthetic health data.

Where privacy of synthetic health data has been evaluated, it has usually been summarised as a single number for a whole cohort. That is precisely the summary that hides the patients who matter. Rare physiology is simultaneously the thing a shared dataset exists to preserve — brittle control, frequent hypoglycaemia, an unusual response to insulin — and the thing most likely to make its owner identifiable. A generator that is safe on average may still expose the people whose data motivated the release, and an aggregate metric will not say so. This is the sparse-population fairness question, and answering it requires attacking individual patients rather than scoring a distribution.

We measure it directly. Using a public multi-study cohort of 506 people, we train a generator on a background population and 26 further generators that each add exactly one target patient, so that a pair of models differs by one person and nothing else, and we ask whether that person’s real recordings lie closer to the release that saw them. We bound the result between a generator that memorises by construction and a permutation floor computed for each released dataset — a floor that is not near zero, and whose omission makes every release look as though it leaks. Across seven generators and two data compositions we find that exposure from glucose alone is close to universal rather than confined to unusual patients, that it is carried by two coordinates of the signal and no others, that removing the model capacity which stores one of them returns exposure to the floor without any cost in fidelity, and that the

same coordinate can be removed at release time by a custodian who cannot retrain, leaving shape-based glycaemic metrics untouched.

2 Results

2.1 Membership signal in synthetic CGM is bounded between a memorisation ceiling and a permutation floor

Each released dataset is attacked through the same paired comparison: for every target patient we hold one generator that saw them and one that did not, and ask whether the patient’s real windows lie closer to the former’s output (Fig. ??). The two generators differ by that one patient and by nothing else. We bound the resulting statistic from both sides before reading any generator’s score.

The upper bound is a generator that memorises by construction, releasing its training windows unchanged. Its median gap is **0.077** on the glucose-only composition and **0.106** with basal insulin, against **0.010** and **0.018** for the strongest learned generator — an order of magnitude larger, and significantly positive in both arms ($p < 0.001$). The statistic therefore responds to memorisation when memorisation is present. It does not, however, separate the atypical arm from the control arm for this generator, and it should not: a model that copies every patient equally memorises the typical and the atypical alike.

The lower bound matters more, because it is the one the literature omits. We recompute the entire pipeline on permuted membership labels, so that no generator saw any target. Under permutation the count of patients recoverable above AUC 0.55 is not near zero: across the twelve released datasets in this study it ranges from **3 to 11 of 26**, with a median of 7. The floor is a property of the released dataset rather than of the study, so it is computed separately for each one. Reported against zero, every generator in Table 1 would appear to leak; reported against its own floor, most do not.

2.2 Membership is recoverable for nearly every patient from glucose alone, and most confidently for those with atypical profiles

We applied the paired design to seven generators across the two data compositions (Table 1). Read as a raw count, the number of patients whose membership is recoverable above AUC 0.55 suggests that every generator leaks: the counts range from 5 to 24 of 26. The permutation floor removes that impression. Once each released dataset is compared against its own floor, the generators separate sharply, and the separation does not track generation fidelity. The strongest published generator in our comparison, DiM-TS, exposes **24 of 26** patients from glucose alone against a floor of **7**. DiffWave, whose fidelity is an order of magnitude worse, sits at an excess of four patients over its floor; Diffusion-TS at -1 ; FourierDiffusion at two. The two best-fitting learned generators in the table occupy opposite extremes of the exposure column.

Exposure at this level is not confined to unusual patients. Of the 24 patients DiM-TS exposes from glucose alone, **13 of 13 are matched controls** and 11 of 13 are

Table 1 Generation fidelity and membership exposure for seven generators on two data compositions. Context-FID is lower-is-better. “Arm AUC” contrasts the atypical arm against the matched control arm; 0.5 is no separation. “Exposed” counts patients above AUC 0.55 out of 26, given against that release’s own permutation floor. The floor, not zero, is the reference.

Generator	Glucose only			Glucose + basal insulin		
	Context-FID	Arm AUC	Exposed	Context-FID	Arm AUC	Exposed
Copy-paste (ceiling)	0.034	0.562	—	0.043	0.680	—
This work	0.038	0.698	10 vs 10	0.055	0.479	5 vs 5
Backbone, unmodified	—	0.515	6 vs 5	0.055	0.598	7 vs 6
DiM-TS	0.095	0.562	24 vs 7	0.159	0.698	15 vs 8
FourierDiffusion	0.100	0.710	7 vs 5	0.194	0.485	6 vs 3
DiffWave	0.368	0.639	12 vs 8	0.491	0.740	13 vs 11
Diffusion-TS	0.422	0.828	7 vs 8	2.058	0.604	8 vs 4

The copy-paste ceiling has no permutation floor computed; it is a construction rather than a fitted model. The unmodified backbone was not scored for fidelity on the glucose-only composition. TODO: close both gaps.

the atypical targets — that is, every typical patient in the design is recoverable, and marginally more reliably than the atypical ones. A study that reported only the count would conclude that atypical physiology confers no additional risk.

The severity of the exposure tells the opposite story. Within the same release the median per-patient AUC is **0.80** in the atypical arm against **0.635** in the matched control arm, where the same quantity computed on permuted membership sits at 0.51 and 0.50 respectively. The most exposed patient, an atypical one, reaches AUC 0.96; the second, also atypical, 0.89. Atypical patients are therefore not exposed *more often* under this composition — they are exposed *more confidently*, and the threshold count is the wrong instrument for seeing it.

TODO — one closing sentence carrying the clinical reading: from a glucose-only release the question is not whether a patient can be recovered but how securely, and the patients recovered most securely are those whose profiles least resemble the cohort’s.

2.3 The membership signal is carried by absolute glucose level and by temporal ordering

To locate the signal we destroyed one coordinate of the released windows at a time and repeated the attack unchanged (Fig. ??). Removing within-hour detail, by averaging to hourly resolution, costs nothing: the strongest leaking release still exposes **23 of 26** patients with a median AUC of 0.692, against 24 and 0.653 untouched. Whatever is being recovered, it is not fine structure.

Two coordinates do carry it. Differencing the windows, which removes absolute level while preserving shape, takes exposure from **24 to 13 of 26**. Sorting each window, which destroys temporal ordering while preserving the value distribution exactly, takes it to **13 of 26** as well. Neither alone reaches the floor.

Destroying both does. With level and ordering removed together, exposure falls to **7 of 26** at a median AUC of 0.487 — the permutation floor for this release is 7 at 0.503. *The two coordinates jointly account for the entire recoverable signal*; nothing measurable remains once they are gone.

The two are not interchangeable with respect to who is protected. Removing temporal ordering leaves the atypical arm separated from the control arm (median AUC 0.600 against 0.476), whereas removing absolute level collapses that separation (0.568 against 0.531). Level is thus the coordinate that carries the specifically atypical part of the signal, which is what motivates the intervention in Section 2.6.

On the two-channel composition, level alone is sufficient: differencing takes exposure from 15 to **6 of 26** against a floor of 8, at or below chance without any need to touch the ordering.

2.4 Adding basal insulin protects typical patients but leaves atypical patients exposed

The second data composition separates the two arms that glucose alone had left indistinguishable by count. Releasing basal insulin alongside glucose reduces the number of exposed matched controls from **13 of 13 to 5 of 13**, while the atypical arm falls only from 11 of 13 to **10 of 13** (Fig. ??). The protection is real but it is not distributed: it accrues almost entirely to the patients who were already typical.

This is the opposite of what the real recordings predict. On real windows, a single day of basal insulin alone identifies its subject **29.3%** of the time against a chance rate of 0.11%, where a day of glucose alone manages 1.5–2.6%. Basal insulin is a programmed schedule — person-specific, stable, and close in character to the absolute level that Section 2.3 identifies as the carrier of the membership signal. By that logic, adding it should raise exposure, not lower it.

The resolution is on the fidelity axis, and it is why the two axes must be read together. Every generator in Table 1 fits the two-channel composition worse than the one-channel composition. The released set is correspondingly noisier, and membership signal is lost with the fidelity. What survives that loss is the signal that was strongest to begin with, which is carried by the atypical patients. Adding a channel that is intrinsically more identifying therefore lowers measured exposure — but it does so by degrading the release, and it leaves the most vulnerable patients where they were.

TODO — Fig. ?? must show all three quantities on one panel: real-data identifiability by channel, measured exposure by arm, and fidelity. Any two of them alone support the wrong conclusion.

2.5 Removing the model capacity that stores absolute level returns membership signal to the permutation floor

Section 2.3 names absolute level as the carrier. We therefore changed the generator so that level cannot be stored per timepoint, rather than adding a penalty that discourages storing it (Fig. ??).

Two changes were made. First, the training task masks whole coordinates instead of scattered cells: contiguous blocks of time, or the level of the whole window. Scattered-cell masking is solvable by interpolating between neighbours, and interpolation is exactly the local correlation in which identity resides. Second, the channels of the latent bottleneck that carry level are pooled along time and broadcast back, so that this part of the representation takes a single value for the entire window and is *structurally unable* to hold a per-timepoint trace. The capacity is removed, not discouraged.

The released sets show no recoverable membership signal. The median gap is **0.0026** on the glucose-only composition and **0.0005** with basal insulin, and in neither composition is it significantly positive in either arm ($p = 0.064$ and 0.682 ; $p = 0.271$ and 0.249) — against $p < 0.001$ for the strongest learned baseline. Per patient, exposure is **10 against a floor of 10** and **5 against a floor of 5**. Fidelity is not paid for this: at Context-FID **0.038** and **0.055** the generator is the best learned model in Table 1 on both compositions, second only to the copy-paste ceiling.

The mechanism behind the masking change can be read directly from training. The task that withholds absolute level should be harder than the original scattered-cell task if the model is genuinely being forced to generate level rather than to copy it. It is, in every setting we trained, but only by **12–24%**. The contiguous-block task is harder by **2–5×**. Most of the effect of this change therefore comes from forcing the model to generate temporal order, not from withholding level, and we report it that way.

TODO — state that the companion group-decorrelation loss cannot constrain the quantity the pooling preserves, since it z-scores along time and thereby removes exactly the window mean that the pooling broadcasts. The pooling does the work.

2.6 The released windows reproduce twenty clinical glycaemic metrics that the strongest baseline distorts

Fidelity measured in an embedding space does not establish that a release is clinically usable. We therefore computed twenty established glycaemic metrics per window and compared their distributions against the real cohort (Table ??).

The released windows reproduce the real distributions across every family of metric. Mean amplitude of glycaemic excursions is **51.22** against a real **51.23**; glucose variability percentage **42.44** against **42.79**; CONGA-1 **41.83** against **41.69**; time in range **74.3%** against **74.0%**; and time below 54 mg/dL **0.55%** against **0.58%**.

The strongest learned baseline does not. It understates short-term variability by a wide margin — mean amplitude **38.28**, a **25%** shortfall, and glucose variability percentage **26.81**, a **37%** shortfall — and it understates exposure to clinically significant hypoglycaemia by **45%** (**0.32%** against **0.58%** of time below 54 mg/dL). A cohort built from that release would look less variable and safer than the patients it was derived from, which is the direction that matters for a dataset intended to support algorithm development.

A release-time intervention for custodians who cannot retrain.

Section 2.3 identified absolute level as the coordinate that carries the specifically atypical part of the signal. A custodian can remove it from an already-trained generator’s

output: publish each window with its own level subtracted, and restore a level drawn by permutation from the levels of the other released windows. The permutation preserves the marginal distribution of levels across the release exactly, while severing which level belongs to which curve.

We measured what this costs clinically, for each of the twenty metrics, as the Kolmogorov–Smirnov distance between the released distribution and the real one (Table ??). The metrics fall into two groups, and the split is by construction rather than by luck.

Metrics defined on shape are *exactly* unchanged, because subtracting a constant from a window does not move them: mean amplitude of glycaemic excursions stays at KS 0.040, CONGA-1 at 0.017, standard deviation at 0.016, MAGE at 0.026, glucose variability percentage at 0.046, and the lability index and average absolute rate of change likewise do not move by a single digit.

Metrics defined on the value scale are destroyed by stripping alone and *restored by the re-offset*. Time below 54 mg/dL moves from KS 0.006 in the untouched release to **1.000** once the level is removed — a complete loss — and back to **0.192** once a level is restored. Time in range goes 0.012, 0.977, **0.126**. Mean glucose goes 0.022, 1.000, and **0.022**: exactly back, because a permutation leaves the marginal untouched. Level-stripping alone is therefore not a usable release; level-stripping with re-offsetting is.

What the re-offset does not restore, and cannot, is the association between a patient’s glycaemic shape and the level they sit at. That association is the privacy gain and it is also a genuine loss of utility: a study asking whether patients with greater variability run higher mean glucose cannot be run on such a release, and the release must say so.

TODO — add the privacy side of this exact transformation, now computing, as a column of Table ?. The numbers in Section 2.3 are for differencing, which is a different transformation, and must not be reused here.

TODO — Table ?: twenty metrics $\times$ (released, level-stripped, re-offset) $\times$ (median, interquartile range, KS, Wasserstein), built from `results/clinical/after_strip.json`.

3 Discussion

Synthetic continuous glucose monitoring data is being released on the premise that a record which was never copied cannot identify anyone. The measurements here do not support that premise, and they locate the failure precisely enough to act on. From a glucose-only release by the strongest generator we compared, membership was recoverable for 24 of 26 patients against a permutation floor of seven — and the recovery was not confined to unusual physiology. Every one of the thirteen matched controls was recoverable. A study reporting only how many patients cross a threshold would have concluded that atypical patients carry no extra risk; the severity says otherwise, with a median per-patient score of 0.80 in the atypical arm against 0.635 in the control arm, and the two most exposed patients both atypical. Atypical patients

are not exposed more often under this composition. They are exposed more confidently, which is the reading a threshold count cannot produce.

What changes that picture is not the model but the data composition. Releasing basal insulin alongside glucose drops the exposed controls from thirteen of thirteen to five, while the atypical arm barely moves. The protection is real and it is unequally distributed: it accrues almost entirely to the patients who were already ordinary. This is worth stating plainly because the intuition runs the other way. On real recordings a single day of basal insulin identifies its subject 29.3% of the time against 1.5–2.6% for glucose, so the channel that protects here is the one that is intrinsically most identifying. The reconciliation is that every generator we tested fits two channels worse than one, the release is correspondingly noisier, and membership signal is lost along with fidelity. Adding an identifying channel lowered measured exposure by degrading the release — which is not a privacy mechanism anyone should adopt deliberately, and which leaves the most vulnerable patients where they were.

The ablation is the part of this work that generalises beyond the generators we happened to test. Removing absolute level and removing temporal ordering each take exposure from 24 patients to 13; removing both takes it to seven, which is the permutation floor for that release. Two coordinates account for the entire recoverable signal, and fine within-hour structure accounts for none of it. The two are not interchangeable in who they protect: with ordering destroyed the atypical arm still separates from the control arm, while with level destroyed that separation collapses. Absolute level is therefore the coordinate carrying the specifically atypical part of the signal, and that is a statement about glucose traces rather than about any particular architecture.

Both interventions we report follow from that. A model whose latent bottleneck is structurally unable to store level per timepoint — capacity removed rather than penalised — produces releases whose membership gap is not significantly positive in either arm under either composition, at the best fidelity of any learned generator we tested. For custodians who cannot retrain, stripping level at release and restoring it by permutation leaves every translation-invariant glycaemic metric exactly unchanged and returns the scale-dependent ones to usable agreement with the real distributions. TODO — position both against prior work on privacy-preserving synthetic health data, differential privacy for generative models, and nearest-neighbour privacy heuristics; citations needed.

Three things constrain what we can claim. First, and most importantly, our own two readings disagree on one cell. On the glucose-only composition the between-arm contrast for our model is 0.698 with $p = 0.045$, which taken alone reads worse than the unmodified backbone at 0.515, while the per-patient count sits exactly on its own permutation floor and the backbone sits one above. With thirteen targets per arm the between-arm statistic takes discrete values and, within a single replicate whose 26 pairs share one background generator, it is a screen rather than a test. **We report one replicate.** The floor-referenced per-patient reading is the one we regard as load-bearing, and the honest position is that a second replicate is required to settle the contrast rather than that the disagreement can be argued away. Second, the intervention removes what is partly the clinically useful signal: mean glucose is both the strongest carrier of identity and a primary clinical quantity, and while the

permutation re-offset restores its population distribution it permanently severs the association between a patient’s glycaemic shape and the level they sit at. A release built this way cannot support a study asking whether more variable patients run higher mean glucose, and must say so. Third, atypical is defined here by one family of distributional metrics, on one cohort, at one window length; subjects unusual in some other sense may carry different risk.

For a custodian preparing a deposit today, the practical readings are these. Report per-patient exposure against a permutation floor computed on the released dataset itself, because against zero every release appears to leak and the comparison is uninformative. Do not treat a lower aggregate leakage number as evidence of protection without checking the fidelity that produced it. And if the release is glucose alone, expect every patient to be recoverable, not only the unusual ones.

4 Methods

4.1 Sex as biological variable

The source release records a single self-reported `gender` field and no separate biological-sex variable. Across the 506 subjects of the cohort it takes the values female ($n = 274$), male ($n = 231$) and non-binary ($n = 1$). This field was not used at any point: it does not enter the definition of atypical profiles, the matching of controls, the composition of the background set, or any model input, and no result reported here is stratified by it. Whether membership risk differs by sex is therefore an open question that this study is not powered to answer — with 13 targets per arm, a sex-stratified contrast would rest on fewer than seven subjects per cell.

4.2 Cohort and data acquisition

Recordings were drawn from a public multi-study continuous glucose monitoring release comprising 1,291 participants across twelve source studies. From these we retained the 506 subjects with at least 30 complete recording days on the glucose channel, giving 6,072 windows. The same subject set is used for both data compositions, so that the one-channel and two-channel results differ only in what is released and not in who is in the cohort.

4.3 Window construction and subject key

Recordings are sampled every five minutes. A window is one gap-free calendar day of 288 consecutive samples; partial and interrupted days are discarded rather than imputed, so no window contains interpolated values. Values are standardised per channel and clipped at five standard deviations; the glucose channel’s constants are a mean of 145.94 mg/dL and a standard deviation of 57.65 mg/dL, and all clinical quantities in this paper are computed after inverting that transformation.

Subject key.

Participant identifiers in the source release are unique only *within* a source study: 241 of 1,291 identifiers (18.7%) recur across two or more studies, and one identifier spans

five. The subject key used throughout is therefore the pair (source study, participant identifier). Keying on the identifier alone merges distinct people into composite subjects, which in a membership design means holding several people in or out as a single unit.

4.4 Definition of atypical glycaemic profiles

Atypical subjects were identified by consensus across thirteen distributional and variability metrics. For each metric, subjects in the top five per cent were flagged; a subject flagged by at least seven of the thirteen metrics was provisionally called atypical. The whole procedure was repeated under four random seeds and only subjects selected under all four were retained, giving 13 atypical subjects. *The threshold of seven votes is fixed for comparability with earlier work on this cohort and is not read off a mode in the vote histogram*, which is unimodal here.

4.5 Control matching

Each atypical subject was paired with one control drawn from the 493 non-atypical subjects and matched on recording duration, since the number of released windows for a target scales with it and nearest-neighbour distance falls with the size of the released set. Matching was exact or near-exact: median relative gap in recording days was below one per cent and no pair exceeded tolerance. Controls were not matched on any glycaemic quantity, so the control arm remains a sample of ordinary physiology rather than a shape-matched twin of the atypical arm.

4.6 Paired membership-inference design

For each of the 26 targets we trained one generator on a background set of 475 subjects (5,741 windows) that contains no target, and one on that background plus the single target. Both members of a pair share the generator, its hyperparameters and its training seed; they differ by one subject and by nothing else. The background generator is shared across all 26 pairs, so the design is symmetric: both arms are membership pairs, and neither is a non-membership control. Each model releases a number of synthetic windows equal to its own training-set size, which is what a custodian releasing a dataset would do.

4.7 Generative models and training budgets

Seven arms were evaluated: a copy-paste construction that resamples its training windows with replacement, four published generators for physiological time series, the unmodified backbone of our own model, and the backbone with the two structural changes of Section 2.5. TODO — name each published generator, its citation, its capacity and its training budget, and state that all arms were trained to the same budget on this cohort.

4.8 Membership statistic

For a target t with real windows R_t , let d_{IN} be the distance from R_t to the release of the generator that saw t , and d_{OUT} the distance to the release of the generator that did not. The statistic is the gap $d_{\text{OUT}} - d_{\text{IN}}$, positive when the target’s own windows lie closer to the release that was trained on them. Distances are Euclidean over the flattened window, reduced across the released set by its minimum and across the target’s windows by their mean. *This variant was fixed on the copy-paste arm and frozen before any learned generator was attacked*, so that the test is not selected on the outcome.

Matching the released-set size.

A generator trained with one additional subject also releases that subject’s windows, roughly 0.1% more than its paired background model. Nearest-neighbour distance falls with the size of the released set, so this alone would bias the gap upwards. Both releases are therefore cut to the same size, at random and with a fixed seed, before any distance is computed.

4.9 Permutation floor

Per-patient membership scores do not have a floor at chance. To measure the floor for each released dataset we destroyed all within-window structure by permuting the samples inside every window independently, then repeated the entire attack unchanged. The resulting count of patients scoring above AUC 0.55 is the floor for that dataset. It is computed separately for every release rather than once for the study, because it depends on the released set’s own size and dispersion.

4.10 Generation fidelity metrics

Fidelity is reported as Context-FID: a TS2Vec encoder is trained on the real half, real and released windows are embedded to full-series representations, and the Fréchet distance between the two representation sets is taken. We also report a discriminative score, the balanced accuracy of a recurrent classifier trained to separate real from released windows, for which 0.5 is indistinguishable. *Scores for a given release are not exactly reproducible across compute jobs*; on byte-identical inputs with identical seeds we observe differences of one to five per cent, so quantities that are compared with one another were computed in a single job.

4.11 Glycaemic utility metrics

Twenty established metrics of glycaemic variability, hypoglycaemia, hyperglycaemia and overall control were computed per window on the inverted mg/dL scale. Because these metrics are defined per recording rather than per cohort, we compare *distributions* rather than means: each metric is summarised by its median and interquartile range across windows, and the distance between a released distribution and the real one is reported as both a Kolmogorov–Smirnov statistic and a Wasserstein distance.

TODO — cite the source of the metric definitions and their discriminative ratios, and note the deviation from its twelve-day windows.

4.12 Statistical analysis

Per-patient membership scores are Mann–Whitney statistics of the two distance distributions, normalised to the unit interval. The between-arm contrast is a Mann–Whitney test of the atypical arm’s gaps against the control arm’s. *All 26 pairs share one background generator, so their gaps are correlated*; within a single replicate the resulting p -values are a screen rather than a test, and the corresponding test is the spread of the arm difference across independent replicates. **This study reports one replicate**, and conclusions that depend on the between-arm contrast alone are qualified accordingly in the Discussion.

Data availability. TODO — must name the deposited synthetic dataset, its repository, its persistent identifier and its licence. The collection requires this at submission.

Code availability. The code used to generate the results presented in this study is publicly available at https://github.com/Cranooooooooo/Metabonet_CGM_MIA.

Acknowledgements. TODO.

Author contributions. TODO.

Funding. TODO.

Competing interests. The authors declare no competing interests.

Supplementary information. TODO.

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